Camunda Gateways and XOR gateway

In this session we will look at what is Inclusive OR gateway and how it functions ?.

- The Inclusive Gateway can be seen as a combination of an exclusive and a parallel gateway. Like an exclusive gateway, you can define conditions on outgoing sequence flows and the inclusive gateway will evaluate them. However, the main difference is that the inclusive gateway can receive more than one sequence flow, like a parallel gateway.

- A diverging Inclusive Gateway (Inclusive Decision) can be used to create alternative but also parallel paths within a Process flow

- Unlike the Exclusive Gateway, all condition Expressions are evaluated

- The true evaluation of one condition Expression does not exclude the evaluation of other condition Expressions

- All Sequence Flows with a true evaluation will be traversed by a token

- Since each path is considered to be independent, all combinations of the paths MAY be taken, from zero to all. However, it should be designed so that at least one path is taken.

- The functionality of the inclusive gateway is based on the incoming and outgoing sequence flows:

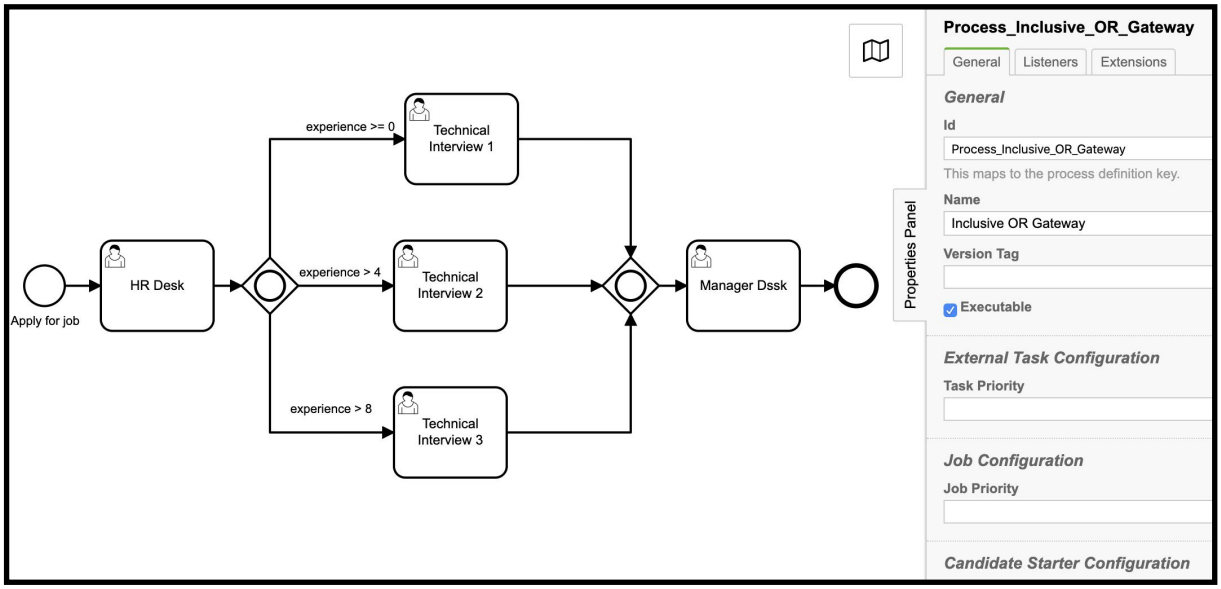
  fork: all outgoing sequence flow conditions are evaluated and for the sequence flow conditions that evaluate to 'true', the flows are followed in parallel, creating one concurrent execution for each sequence flow.

  join: all concurrent executions arriving at the inclusive gateway wait at the gateway until an execution has arrived for each of the incoming sequence

flows that have a process token. This is an important difference to the parallel gateway. So in other words, the inclusive gateway will only wait for the incoming sequence flows that are executed. After the join, the process continues past the joining inclusive gateway.

1. **We will model below process to demonstrate use of inclusive gateways.**

HR Desk, will fill up candidate information from resume and complete the task. Based on the experience of candidate that many interviews will be scheduled. Eg. If candidate experience is 6 years then user tasks for Technical Interview 1 and Technical Interview 2 will be created. Once these interviews are completed then Manager Desk task is created with all the information of candidate and technical interviews evaluation.

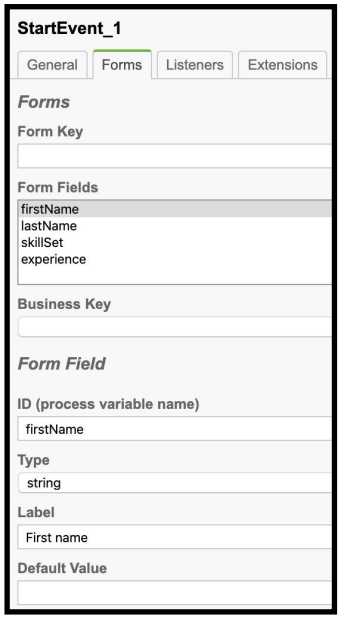


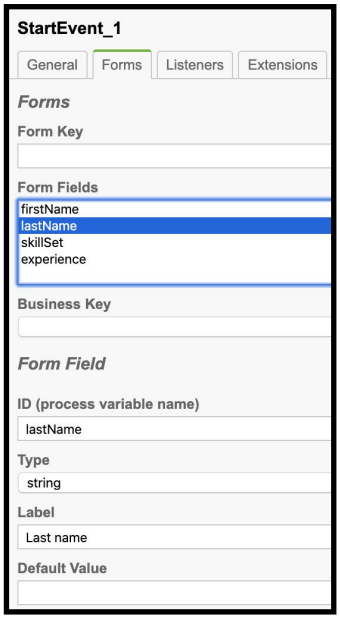
Process properties:

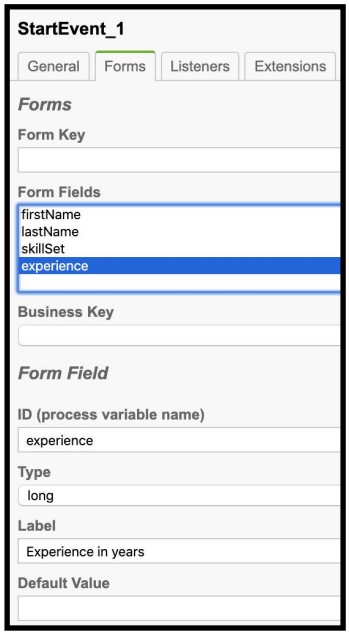
Id: Process_Inclusive_OR_Gateway

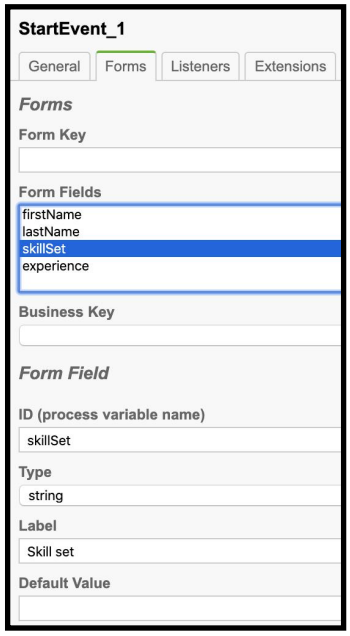
Name: Inclusive OR Gateway

2. Start event: Apply for job

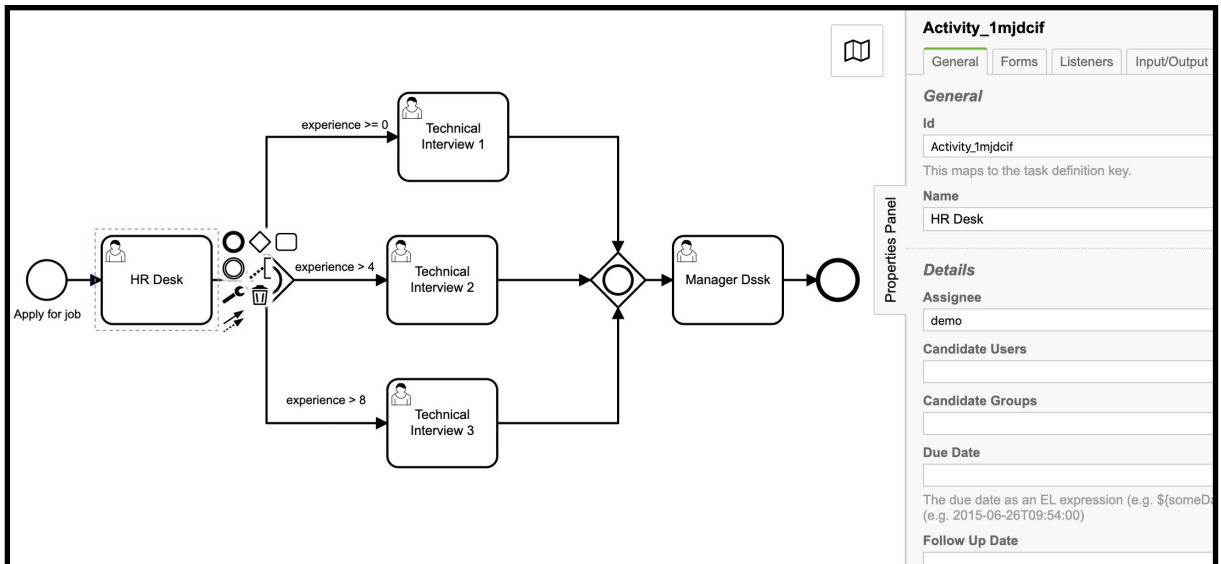








3. User task: HR Desk



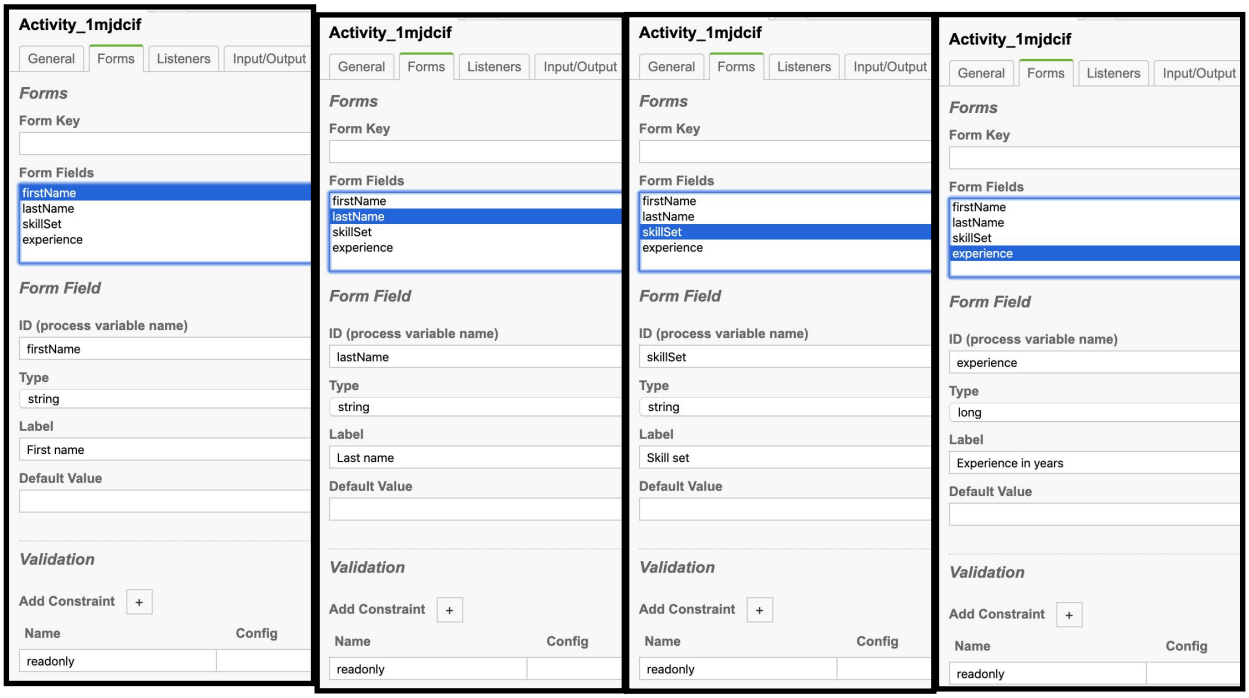
Properties:

Id: Activity_1mjdcif

Name: HR Desk

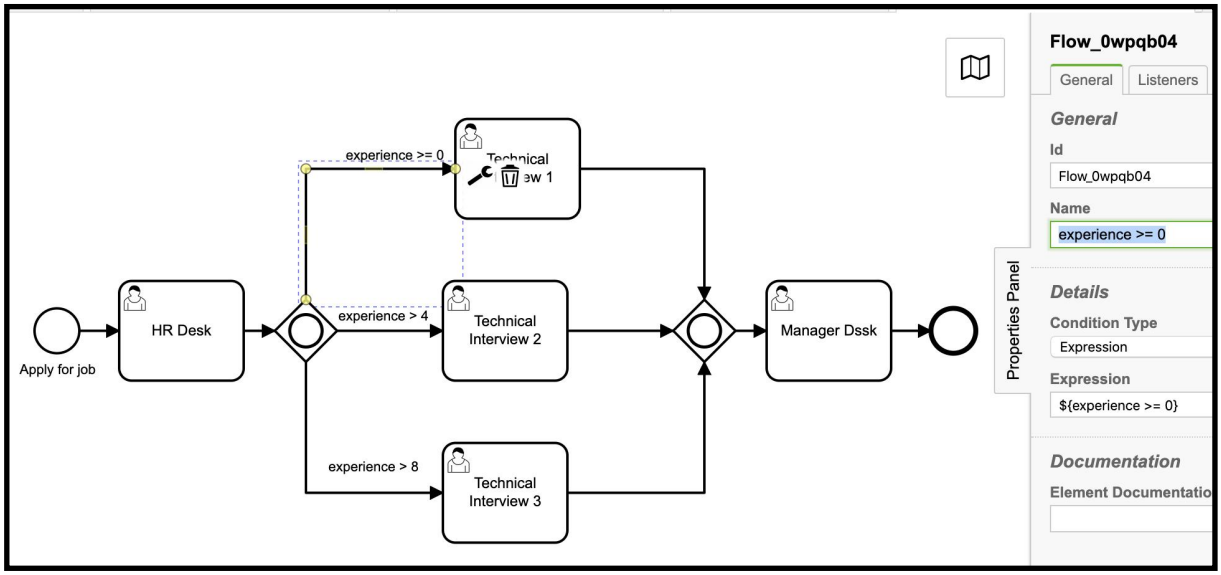
Assignee: demo

Task form:

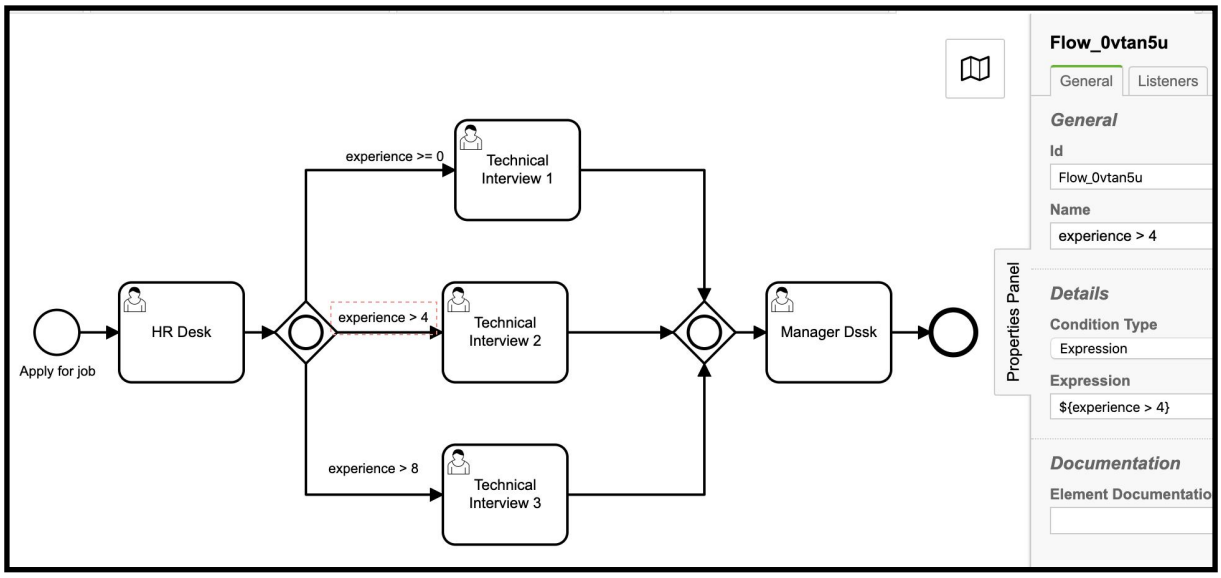


4. Sequence flow : Constraints

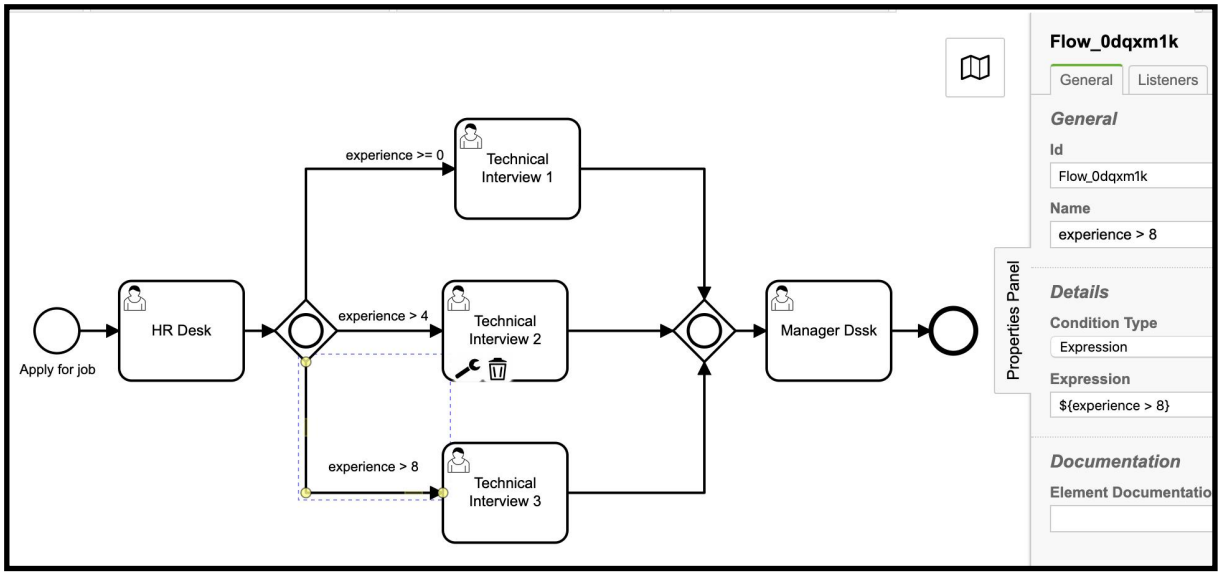
experience >= 0



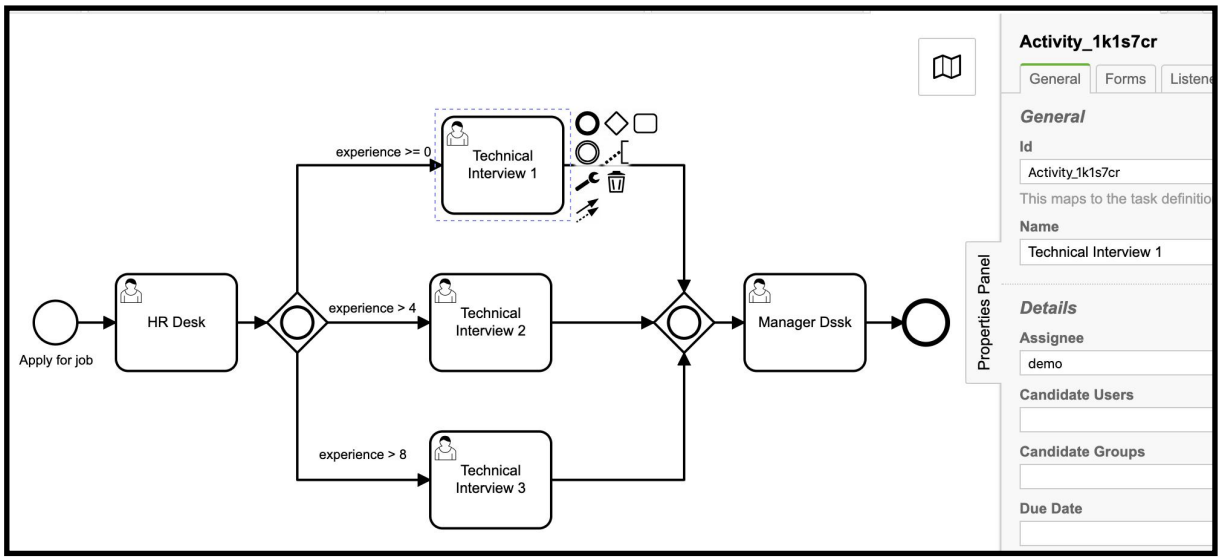
experience > 4



experience > 8



5. User task: Technical Interview 1

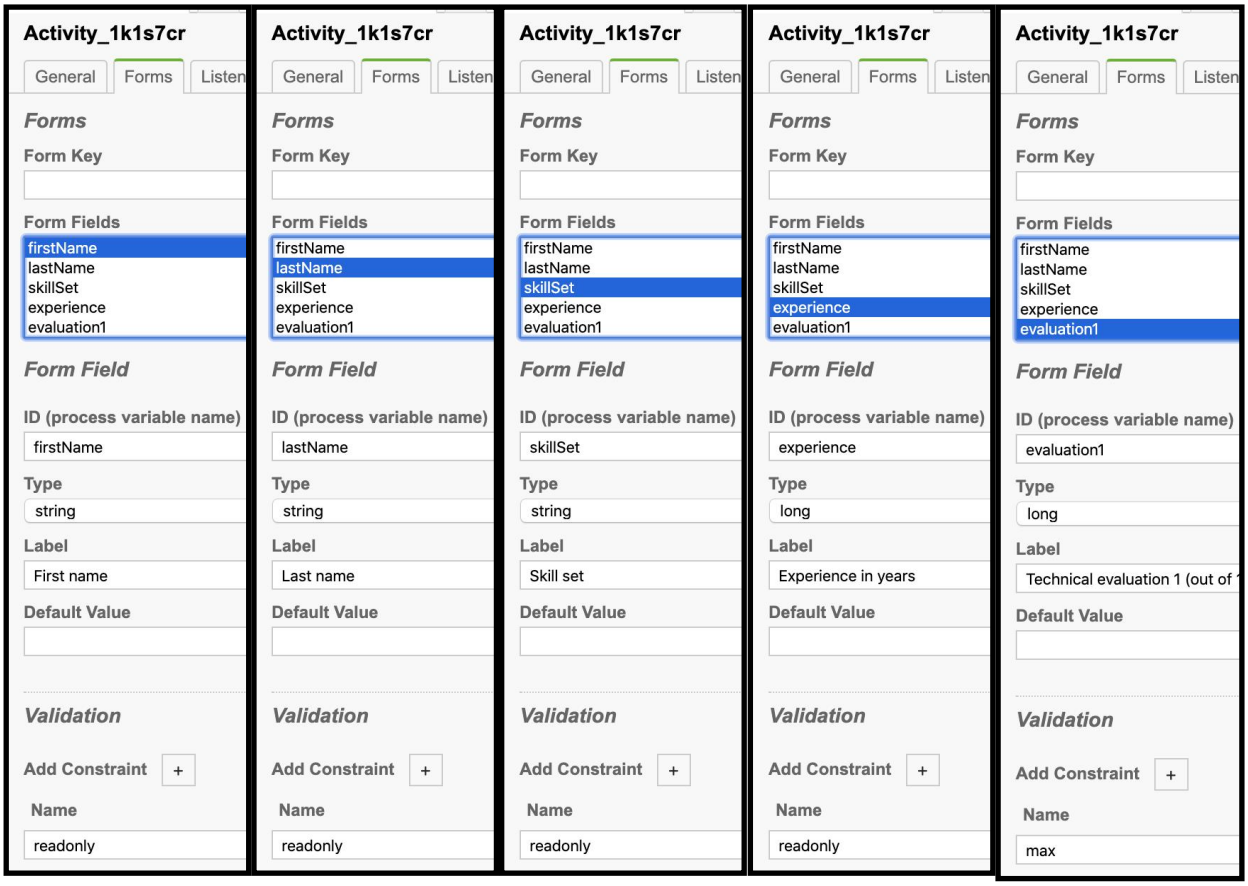


Id: Activity_1k1s7cr

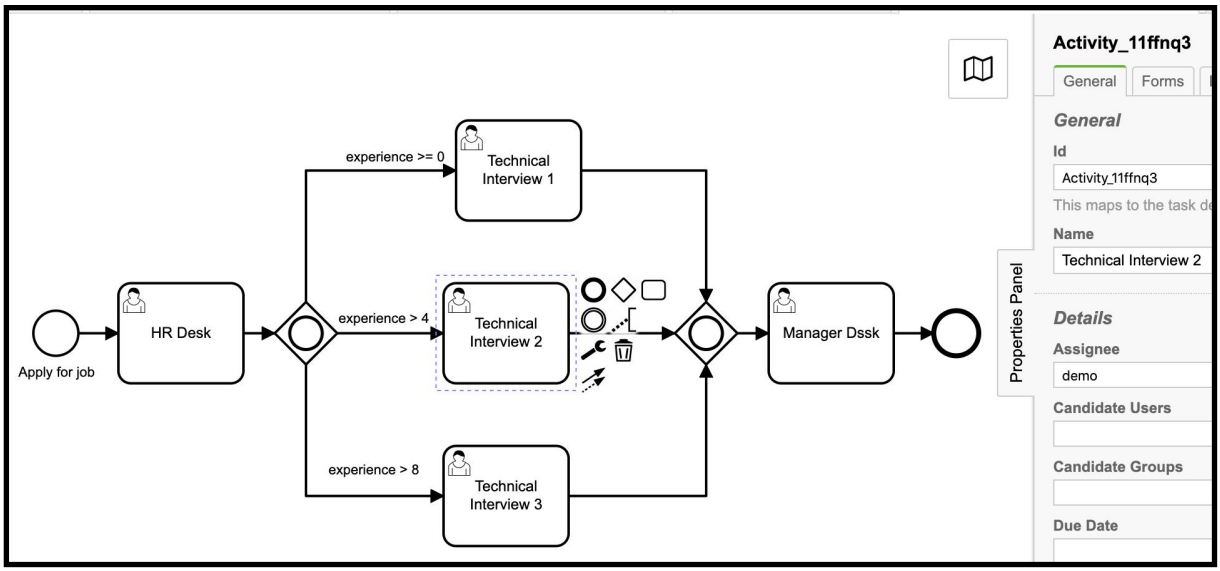
Name: Technical Interview 1

Assignee: demo

Task form:



6. User task: Technical Interview 2

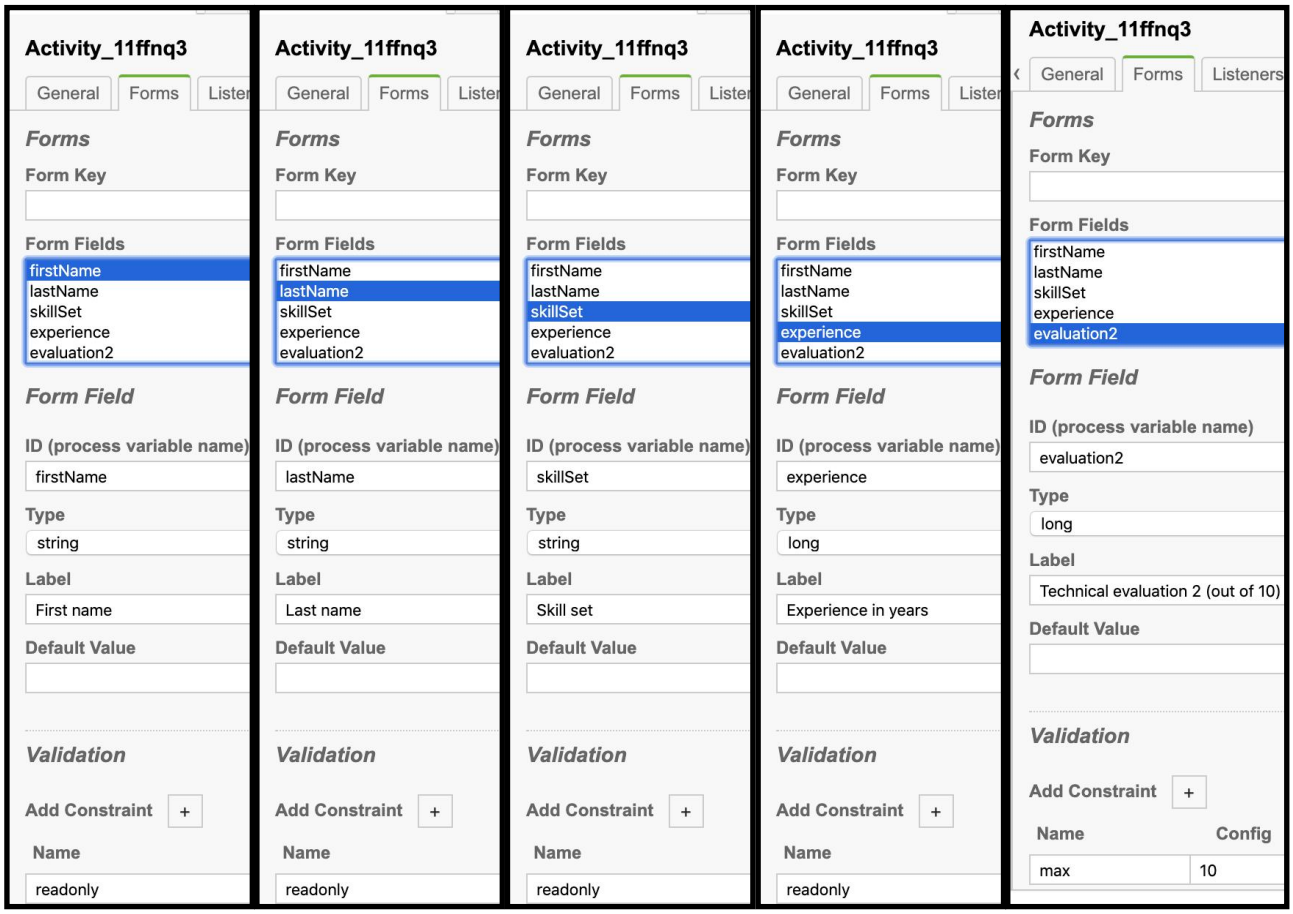


Id: Activity_11ffnq3

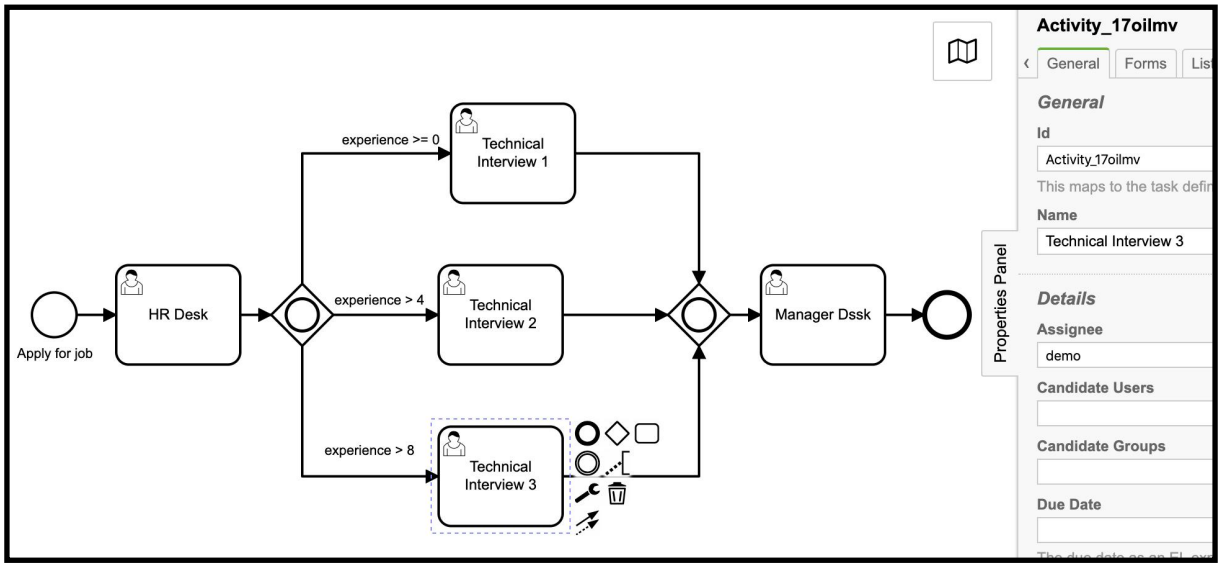
Name: Technical Interview 2

Assignee: demo

Task form:



7. User task: Technical Interview 3

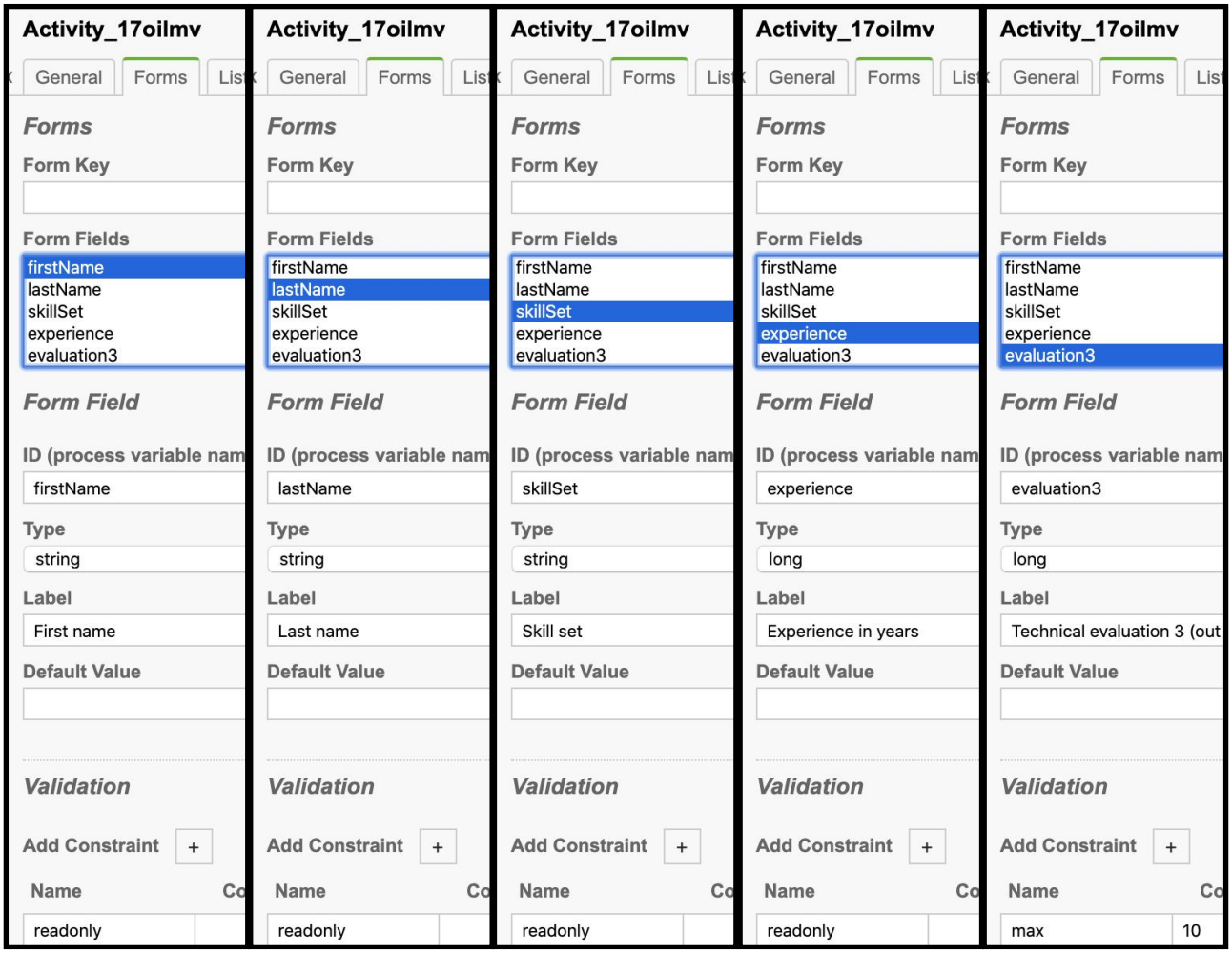


Id: Activity_17oilmv

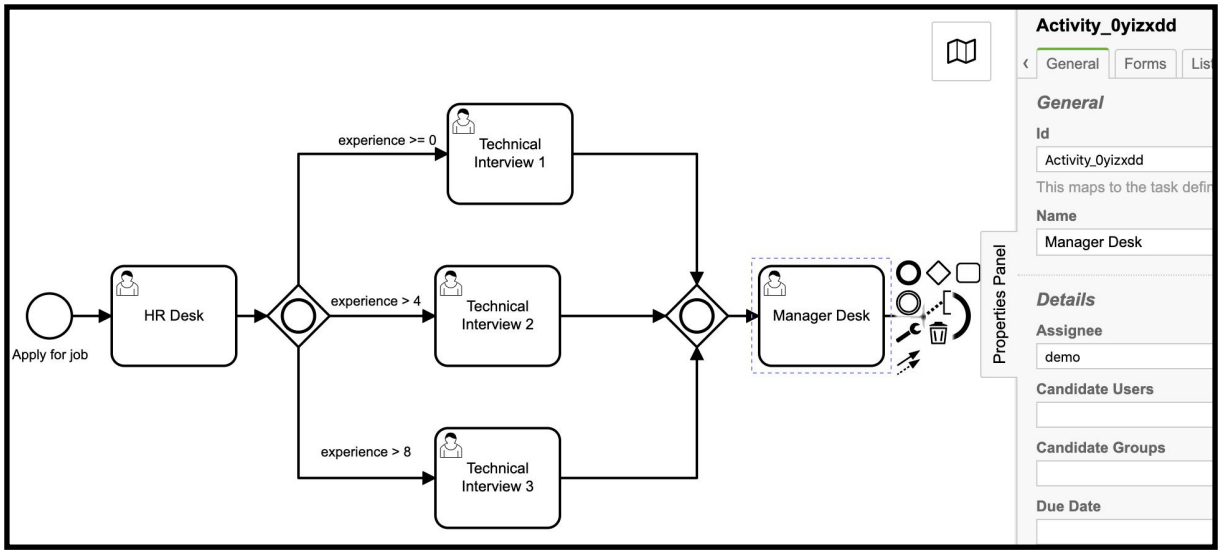
Name: Technical Interview 3

Assignee: demo

Task form:



8. User task: Manager Desk

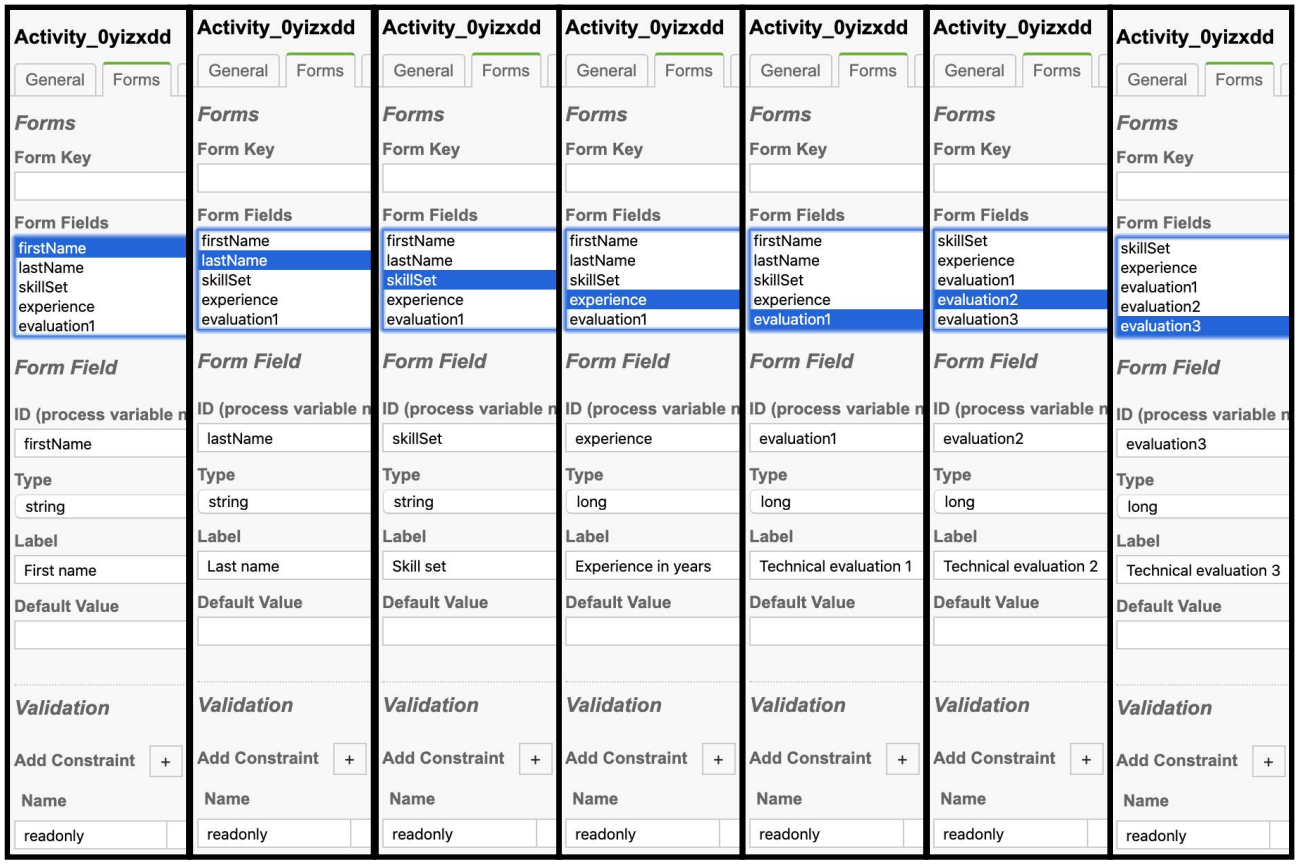


Id: Activity_0yizxdd

Name: Manager Desk

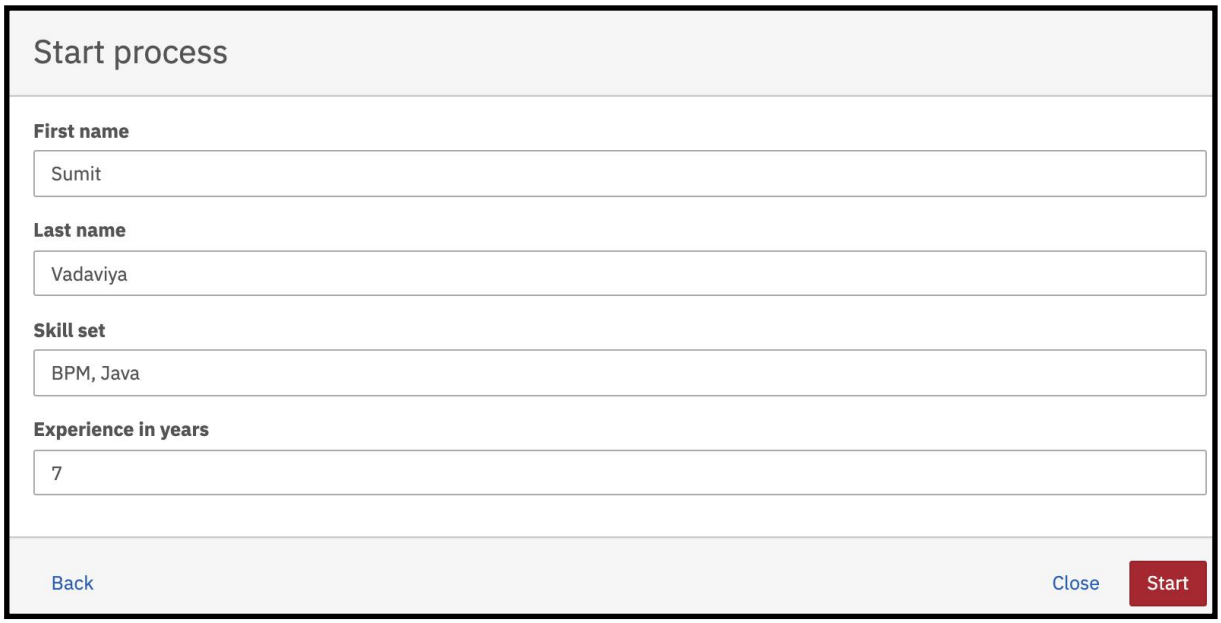
Assignee: demo

Task form:



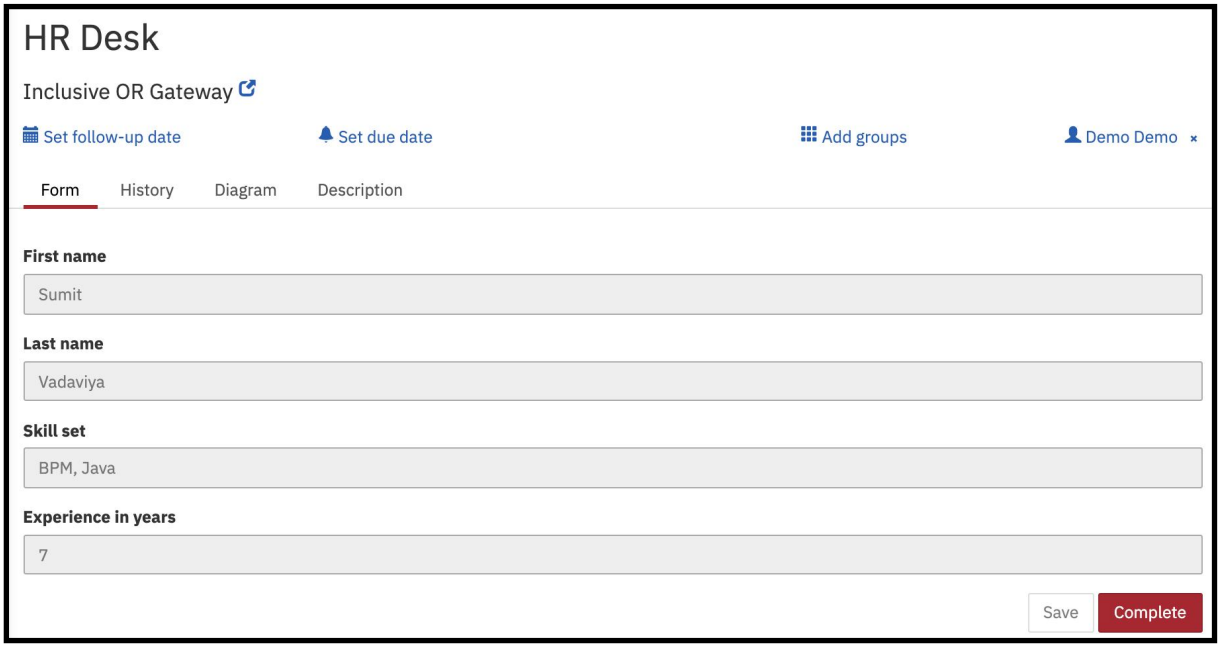
9. Deploy and start the process

As per process definition start event form for apply for job will appear as per below.



Hit the Start button. As per process definition user task HR Desk will get created and assigned user is demo.

Open HR Desk task

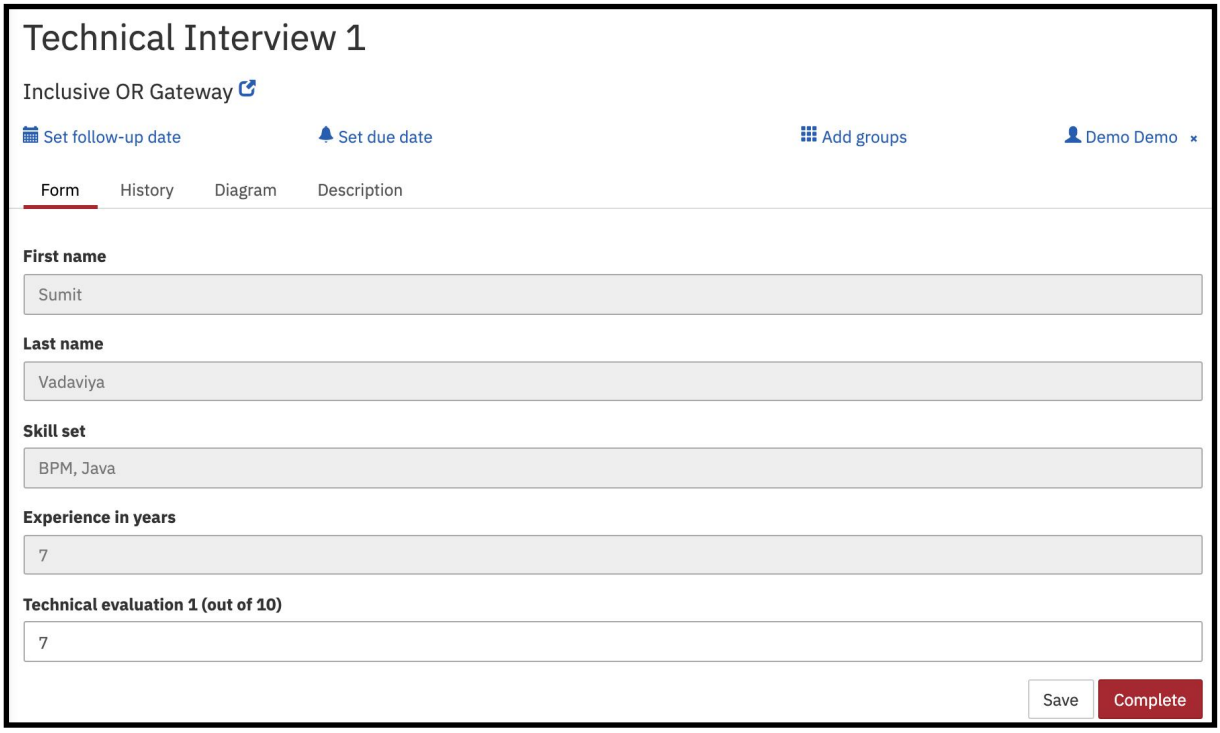


Hit the Complete button.

Now as we have used Inclusive OR gateway and based on input

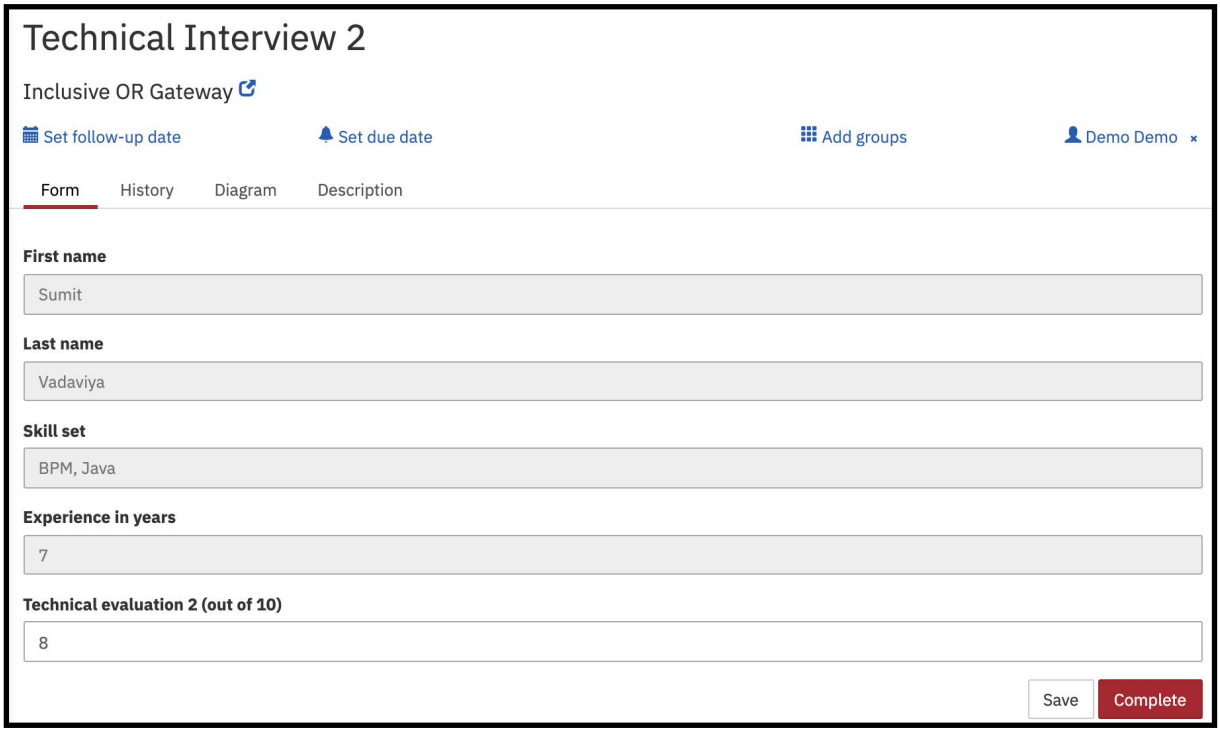
For **Experience in years = 7**, first two sequence flow constraints(experience >= 0 and experience > 4) will get satisfied and as a result user tasks **Technical Interview 1** and **Technical Interview 2** will get created.

Open user task Technical Interview 1 and complete



User has input Technical evaluation 1 : 7. Hit the complete button.

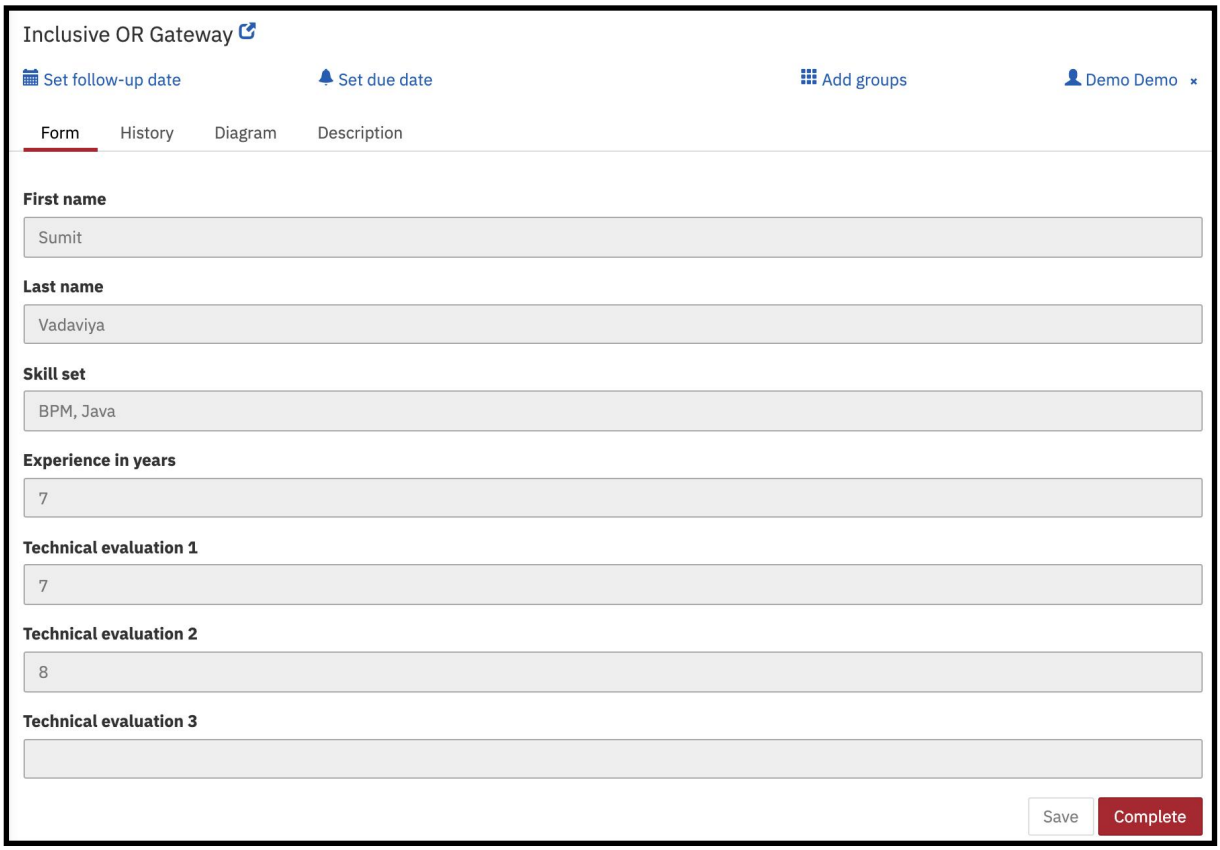
Open user task Technical Interview 2 and complete



User has input Technical evaluation 2 : 8. Hit the complete button.

Now as per process definition user task Manager Desk will get created.

Open user task Manager desk and complete



As you can see manager will see technical evaluations provided by previous tasks and finally hit the Complete button to stop the process.